

CHE 435/525 *Process Systems Analysis and Control*

Lecture 2: **Dynamic Modeling Principles & Simple Examples**

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Discuss & Recapture

- What is feedforward and feedback control?
- What is a controlled/manipulated/disturbance variable (CV/MV/DV)?
- How do you state a control objective?
- Growth mindset v.s. fixed mindset



© Big Change

Learning objectives of this lecture

- ① Apply the general principles of building a **dynamic** process model
- ② Model typical **tank** systems: one-component/multi-component, adiabatic/heated
- ③ Perform **degree-of-freedom (DoF)** analysis on dynamic models

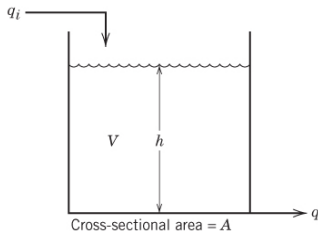
Principle 1: Conservation Laws

- Accumulation rate = In – Out + Generation – Consumption
- Mass balances:
 - ▶ Total balance and component balances
 - ▶ Generation/consumption terms in reactions: Use reaction rate laws
- Energy balance:
 - ▶ Recall from thermodynamics

$$\frac{dU_{\text{tot}}}{dt} = \sum_{j:\text{streams}} w_j \hat{H}_j + \dot{Q} + \dot{W}_{\text{shaft}}.$$

- ▶ w_j : mass/molar flow rate, $\hat{H}_j$: specific/molar enthalpy, U : internal energy

Example 1: Liquid storage tank (isothermal)



(q_i and q : volumetric flow rates)

- Mass balance:

$$\frac{dm}{dt} = \rho q_i - \rho q \quad \Rightarrow \quad \frac{dh}{dt} = \frac{q_i - q}{A}.$$

- q_i : a constant, or a DV, or an MV (depending on the actual setting)
- q : either an MV (if a valve is installed to manipulate), or related to h :

$$q = h/R_v \quad \text{or} \quad q = C_v \sqrt{h}$$

Which one is more accurate? You may do an experiment.

Example 1 (cont.): Liquid storage tank (isothermal)

- Assumptions:

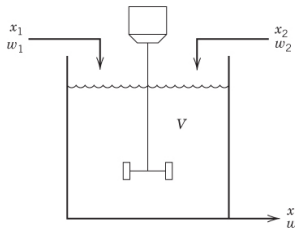
- 1 q_i is a constant;
- 2 C_v (valve constant) and ρ (density) are constant parameters;
- 3 $q = C_v\sqrt{h}$: the out-flow from the bottom is naturally pressed by the liquid in the tank.

- Model:

$$\frac{dh}{dt} = \frac{q_i - C_v\sqrt{h}}{A}.$$

- ▶ An **ODE** describing how the liquid level changes with time
- ▶ Can be solved under given initial condition $h(0) = h_0$

Example 2: Blending tank with variable holdup



(w_1, w_2, w : mass flow rates)

- Mass balances:

$$\frac{d(\rho V)}{dt} = w_1 + w_2 - w$$

$$\frac{d(\rho V x)}{dt} = w_1 x_1 + w_2 x_2 - w x$$

Example 2 (cont.): Blending tank with variable holdup

- Assumptions:

- 1 w_1 and x_2 are constants;
- 2 x_1 is a DV;
- 3 w_2 and w are MVs;
- 4 ρ is a constant, not affected by x (How do you like it?)

- Model:

$$\frac{dV}{dt} = \frac{1}{\rho}(w_1 + w_2 - w)$$

$$\frac{d(Vx)}{dt} = \frac{1}{\rho}(w_1x_1 + w_2x_2 - wx)$$

- Simplification of the second equation:

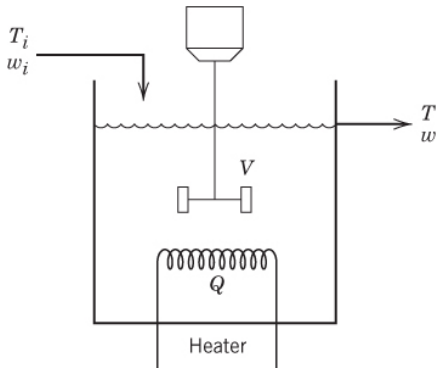
$$d(Vx)/dt = V \cdot dx/dt + x \cdot dV/dt$$

$$\frac{dx}{dt} = \frac{w_1}{\rho V}(x_1 - x) + \frac{w_2}{\rho V}(x_2 - x).$$

Principle 2: Constitutive equations

- Use thermodynamics, transport, reaction kinetics to relate variables
- When heat effect exists:
 - ▶ Relate $\hat{U}$, $\hat{H}$ to (T, P, x)
 - ▶ Use heat capacity and mixture models (ideal mixture or fugacity equations)

Example 3: Heating Tank



- Assumptions:

- 1 w_i is a DV, w an MV
- 2 T_i is a DV, Q an MV
- 3 $C_P = \text{const.}$, $\rho = \text{const.}$

Example 3 (cont.): Heating Tank

- Mass balance:

$$\frac{dV}{dt} = \frac{w_i - w}{\rho}$$

- Energy balance:

$$\frac{dU_{\text{tot}}}{dt} = w_i \hat{H}_i - w \hat{H} + Q$$

- ▶ $\hat{H}_i = C_P T_i$, $\hat{H} = C_P T$ (assuming no phase change)
- ▶ $U_{\text{tot}} = \rho V \hat{U} \approx \rho V \hat{H} = \rho V C_P T$ (assuming $\hat{H} \approx \hat{U}$ for liquid)

- Simplification of energy balance:

$$\begin{aligned} \rho \left(C_P T \frac{dV}{dt} + V C_P \frac{dT}{dt} \right) &= w_i C_P T_i - w C_P T + Q \\ \Rightarrow C_P T (w_i - w) + \rho V C_P \frac{dT}{dt} &= C_P (w_i T_i - w T) + Q \\ \Rightarrow \frac{dT}{dt} &= \frac{w_i}{\rho V} (T_i - T) + \frac{Q}{\rho V C_P} \end{aligned}$$

Once you have a model, confirm with DoF Analysis!

- ① List known quantities / parameters as constants.
- ② Determine # equations (N_E) and # process variables (N_V).
 - ▶ Time t is not considered as a process variable.
- ③ Calculate $N_F = N_V - N_E$ as # DoFs.
- ④ Identify N_E output variables to be solved from ODEs.
- ⑤ There must be exactly N_F input variables (DVs/MVs). Identify them.

DoF for the Heating Tank Example

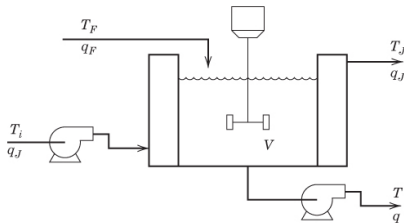
- Model:

$$\frac{dV}{dt} = \frac{w_i - w}{\rho}$$

$$\Rightarrow \frac{dT}{dt} = \frac{w_i}{\rho V} (T_i - T) + \frac{Q}{\rho V C_P}$$

- Constants: ρ, C_P
- $N_E = 2$ and $N_V = 6$ (V, T, w_i, w, T_i, Q)
- $N_F = 4$ DoFs.
- 2 outputs: V, T . They are dependent on ...
- 4 inputs: 2 MVs (w, Q), 2 DVs (w_i, T_i)

Example 4: Jacket-heated tank [Seborg Ex.2.8]



- Assumptions:

- 1 Tank holdup V and jacket holdup V_J are constants
- 2 Heat loss from the vessel is negligible
- 3 Tank contents and jacket contents have **significant** thermal capacitances
- 4 Tank wall and jacket wall have **negligible** thermal capacitances
- 5 Heat transfer coefficient [Btu/h/ft²/degF] $U = Kq_J^{0.8}$ ($K = \text{const.}$)

- Model: **"Two-tank"**

Example 4 (cont.): Jacket-heated tank

- Energy balance for the tank

$$\frac{dT}{dt} = \frac{q}{V}(T_F - T) + \frac{Q}{\rho V C_P}$$

where $Q = UA(T_J - T) = Kq_J^{0.8}A(T_J - T)$

- Energy balance for the jacket

$$\frac{dT_J}{dt} = \frac{q_J}{V}(T_i - T_J) - \frac{Q}{\rho_J V_J C_{PJ}}$$

- **Engineers' insight: How can you model fluently?**

- 1 Understand the meaning of each term.
- 2 Verify dimensions (units).
- 3 Use analogy.